Princess Sumaya University for Technology - King Hussein School for Computing Sciences

CS11313 - **Algorithms Design and Analysis** - **Spring** 2022

Quiz # 2

Name:	**ID:**

Q. Solve each of the following recurrence equations using the Master Method. If the Master Method does not apply, explain briefly why and then solve the equation using iterative substitution ***or*** the recursion tree method.

A. $T(n) = 8T(\frac{n}{2}) + n^2 \log_2 n, \quad T(1) = 0$

Work at the root = $n^2 \log_2 n$.
Number of leaves = $n^{\log_2 8} = n^3$.

$n^2 \log_2 n = O(n^{3-\epsilon})$ for $0 < \epsilon < 1$. Therefore:

$T(n) = \Theta(n^3)$.

B. $T(n) = T(\frac{n}{3}) + \log_3(n), \quad T(1) = 0$

Work at the root = $\log_3 n$.
Number of leaves = $n^0 = 1$.

$\log_3 n \neq O(n^{0-\epsilon})$
$\log_3 n \neq \Theta(1)$
$\log_3 n \neq \Omega(n^{0+\epsilon})$

Since there is no polynomial separation between the work at the root and the number of leaves, the Master Method can't be applied.

<u>Using the Recursion Tree Method</u>

Height = $\log_3 n$.
Work at level *i* = $\log_3(\frac{n}{3^i})$

Total = $\sum_{i=0}^{\log_3 n} \log_3(\frac{n}{3^i}) = \sum_{i=0}^{\log_3 n} (\log_3 n - \log_3(3^i)) = \sum_{i=0}^{\log_3 n} \log_3 n - \sum_{i=0}^{\log_3 n} i \log_3 3$

Total = $(\log_3 n(\log_3 n + 1)) - (\frac{1}{2} \log_3 n(\log_3 n + 1)) = \Theta(\log^2 n)$

Math Cheatsheet

A. Logarithms

1. $a^b = c \rightarrow b = \log_a c$ ← Definition
2. $\log_b b = 1,\ \log_b 1 = 0$ ← Special cases
3. $\log_b(\frac{x}{y}) = \log_b x - \log_b y$
4. $\log_b(x \times y) = \log_b x + \log_b y$
5. $\log_b x^y = y \times \log_b x$ ← Follows directly from the previous rule.
6. $\log_b x = \dfrac{\log_c x}{\log_c b}$ ← Changing bases
7. $x^{\log_b y} = y^{\log_b x}$
8. $b^{\log_b x} = x$ ← Follows directly from the previous rule.
9. $\lg(n!) \sim n \lg n$ ← Stirling's Approximation

Notation:

- $\log n$ (no base) → used with orders of growth to indicate that the base is not important. Logarithms with different bases differ by a constant factor as shown in the listed identities.
- $\lg n$ → base 2.
- $\ln n$ → natural logarithm (base is e)

B. Summations

1. $\sum_{i=1}^{n} i = 1 + 2 + \ldots + n$ ← Definition
2. $\sum_{i=1}^{n} c = c + c + \ldots + c = c \times n$ ← If c does not depend on i.
3. $\sum_{i=1}^{n} c \times f_i = c \times \sum_{i=1}^{n} f_i$
4. $\sum_{i=1}^{n} f_i + g_i = \sum_{i=1}^{n} f_i + \sum_{i=1}^{n} g_i$
5. $\sum_{i=1}^{n} i = 1 + 2 + 3 + \ldots + n = \dfrac{n(n+1)}{2}$
6. $\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$
7. $\sum_{i=0}^{n} r^i = r^0 + r^1 + r^2 + \ldots + r^n = \dfrac{r^{n+1} - 1}{r - 1}, \quad r \neq 1$ ← Geometric Sum.
8. $\sum_{i=0}^{n} 2^i = 2^{n+1} - 1\ , \qquad \sum_{i=0}^{n} (\frac{1}{2})^i = 1 + \frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^n} \sim 2$ ← Special cases of a geometric sum ($r = 2$ and $r = 0.5$).
9. $\sum_{i=1}^{n} \dfrac{1}{i} = 1 + \dfrac{1}{2} + \dfrac{1}{3} + \ldots + \dfrac{1}{n} \sim \int_1^n \dfrac{1}{i}\, \mathrm{d}i = \ln n$ ← Harmonic Number H_n.